

A Knudsen-extended, calendering-aware Zehner–Bauer–Schlünder closure for the effective thermal conductivity of lithium-ion electrode coatings and separators

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Abstract

Battery thermal models need the effective through-plane thermal conductivity λ_{eff} of every porous layer in the cell, electrode coatings and separator alike. In practice this quantity is taken from closures, compact analytical formulas that predict the effective conductivity of a porous medium from its constituent properties and structure; the closures in use know only the porosity, treat the pore gas as a continuum, and come without an uncertainty statement. We develop and test a closure that repairs all three deficiencies, and we explain along the way why each repair matters. First, a Smoluchowski/Knudsen correction accounts for the fact that gas in very small pores conducts heat poorly: the effect is a systematic 3% to 6% for electrode coatings measured by laser flash in helium, and it removes 84% of the pore-gas conduction in polyolefin separators, whose 43–64 nm pores are narrower than the mean free path of air. Second, a compression-dependent contact term $\varphi(\Pi)$ represents what calendering actually does to a coating: it first damages and later rebuilds the network of conductive bridges between particles. Third, the closure is implemented end-to-end differentiable, so it can be run backwards, turning a thermal measurement into a manufacturing quality-control reading. Tested against the open calendering dataset of Gandert et al. (4 electrode types $\times$ 27 calendering states), the model with zero fitted parameters predicts the electrode that best matches its assumptions (uncalendered NMC811) to +2.7% and outperforms the classical effective-medium baselines (Wiener bounds, Maxwell–Eucken, Bruggeman). An ablation study shows the mean error falling from 31.1% over 13.5% to 4.5% as the contact physics is added, reaching the noise floor of the measurements and reproducing, for the first time to our knowledge, the experimentally observed u-shaped dependence of conductivity on calendering degree. An identifiability and transfer analysis shows that the failure to transfer a calibration between electrode recipes is carried almost entirely by a single, physically meaningful parameter (the as-coated bridge conductance, set by the conductive-additive recipe), which yields concrete material design rules. The inverse mode recovers coating porosity from one noisy thermal measurement to ± 0.008 absolute and offers the first candidate inline monitor for interface and delamination quality. A validation campaign is specified, centered on a two-gas-pressure experiment that cleanly separates pore-gas conduction from solid-skeleton conduction.

Keywords: effective thermal conductivity; lithium-ion battery; calendering; Knudsen effect; effective-medium theory; inline quality control

1 Introduction

Thermal management decides how long a lithium-ion battery lives and how safely it operates. Elevated temperatures accelerate the side reactions that consume capacity, temperature *gradients* across a cell make its regions age at different rates, and in the worst case a local hot spot can trigger thermal runaway [1, 2]. Designing against all three requires thermal *models*, and every such model needs the effective through-plane thermal conductivity λ_{eff} of each layer in the stack as an input. Through-plane conduction is the bottleneck: cell-level cross-plane values of $0.15\text{--}0.8\text{ W m}^{-1}\text{ K}^{-1}$ lie roughly two orders of magnitude below the in-plane values [2, 3], so heat generated in the interior must cross many poorly conducting porous layers before it can leave the cell.

Moreover, λ_{eff} depends strongly on processing. Calendering, the roll-pressing step that compacts a coated electrode to its target porosity, changes the conductivity substantially: in the measurements of Gandert et al. [4], calendering a graphite anode from porosity $\varepsilon = 0.60$ down to $\varepsilon = 0.21$ raised the measured electrode conductivity from about 1.5 to $2.4 \text{ W m}^{-1} \text{ K}^{-1}$, a change of roughly 60%. A thermal model that treats λ_{eff} as a handbook constant therefore embeds a hidden, process-dependent error which, once wetting state and porosity are both accounted for, approaches an order of magnitude.

Decades of research on heat conduction in packed beds and porous media have produced closures, compact formulas that predict λ_{eff} from the constituent conductivities and the pore structure, and the Zehner–Bauer–Schlünder (ZBS) family [8, 9, 10] is the most widely used of them, including for battery components [5, 4]. Yet recent measurements expose a structural mismatch between what these closures describe and what electrode manufacturing produces. Three gaps stand out.

(i) *The pore gas is not a continuum.* Electrode pores measure a few micrometers and separator pores only tens of nanometers, while an air molecule travels on average 67 nm between collisions at ambient conditions [16]. In such pores, gas conducts heat far worse than the handbook value suggests (Sec. 3.3 quantifies this). The effect is well known in insulation research, but it has not been built into ZBS-based battery modelling; Gandert et al. [4] explicitly flag it as a known but unquantified correction in their own data evaluation.

(ii) *Porosity alone cannot describe calendering.* Gandert et al. measured four electrode types across 27 calendering states and found a *non-monotonic, u-shaped* dependence of λ_{eff} on porosity, which correlates with the adhesion between coating and current collector. They compared two state-of-the-art models, an analytic homogenization closure [5] and a discrete-element simulation [6], and reported poor agreement: both rise monotonically as porosity falls, because both see only the pore volume, while calendering acts on the *contacts* between particles.

(iii) *Closures are only ever run forwards.* Their potential as *inverse* tools, reading porosity or interface quality back from a fast thermal measurement, is unexploited, although interface quality dominates the thermal resistance of a cell stack [7] and currently has no inline monitor at all.

This paper closes these gaps with a deliberately minimal extension of the ZBS closure, validated and calibrated on openly published data. Our contributions are:

- C1 A Knudsen-extended ZBS closure** for battery components (Sec. 3.3). We quantify a correction that has previously been noted but not evaluated for battery components: 3% to 6% for electrodes under laser-flash conditions (helium atmosphere), and -84% of pore-gas conduction for polyolefin separators, where we additionally show that classical continuum sphere-pack models match measurements only through a cancellation of two errors (Sec. 5.2).
- C2 A calendering-aware contact term $\varphi(\Pi)$** (Sec. 3.4), to our knowledge the first closure that reproduces the measured u-shape. An ablation (Sec. 5.3) attributes the error reduction, $31.1\% \rightarrow 13.5\% \rightarrow 4.5\%$, to its individual components; the endpoint is the dataset’s own noise floor.
- C3 An error, identifiability, and transfer analysis** (Sec. 5.5). A profile-likelihood study bounds which parameters the data can and cannot determine; hybrid-swap experiments show that cross-recipe transfer failure is carried by one parameter, the as-coated bridge conductance, giving a one-measurement re-anchoring rule and volume-space design principles for additive and binder selection (Sec. 5.4).
- C4 Differentiable inverse tools for manufacturing quality control** (Sec. 5.6): coating porosity from a single 3%-noise thermal reading to ± 0.008 absolute with a calibrated uncertainty, and a bridge/core decomposition that makes thermal metrology the first candidate inline monitor for interface and delamination quality.
- C5 A falsifiable experimental program** (Sec. 7), centered on a two-gas-pressure protocol that separates pressure-dependent pore-gas conduction from pressure-independent skeleton conduction, testing the mechanism rather than the fit.

All code, data transcriptions, and analysis notebooks are openly available at <https://github.com/j-stoerk/zehner-electrode-thermal> (Reproducibility Statement).

2 Related work

Analytic effective-medium closures. The classical results bracket what any two-phase mixture can conduct: the Wiener bounds [11] are the exact best and worst cases (all material in parallel, or all in series), tightened by the Hashin–Shtrikman bounds [12] for isotropic mixtures. Structure-specific models

then commit to a particular geometry. Maxwell–Eucken [13, 14] pictures isolated particles floating in a connected fluid; Bruggeman’s effective-medium theory [15] treats both phases as interpenetrating; the ZBS family [8, 9, 10] resolves a packed-bed unit cell in which particles touch at points (or, in Bauer’s extension, over small flattened contacts). All of these evaluate in microseconds, differentiate cleanly, and have parameters one can reason about, which is why they dominate engineering practice. Their weakness is the flip side of their strength: each is only as good as its geometric commitment. We turn this weakness into a diagnostic in Sec. 5.2, where the *pattern* of baseline failures (Bruggeman matching co-continuous separators but overshooting sphere-packed electrodes by up to +465%, Maxwell–Eucken undershooting everywhere) identifies the governing microstructure of each component. None of these closures, as used for batteries, includes rarefied-gas corrections or any process variable beyond porosity.

Microstructure-resolved simulation. Discrete-element models [6] and homogenization on real or virtual microstructures [5] represent particle arrangements explicitly, and can in principle resolve contact areas. In practice, both approaches were reduced to porosity-parameterized predictions when compared against the calendering data, and both missed the u-shape [4]: their structure generators do not encode how a calender damages and rebuilds contacts. Resolved simulation also costs orders of magnitude more per evaluation than a closure and does not differentiate conveniently, which forecloses the inverse applications of Sec. 5.6. Our approach is complementary rather than competitive: keep closure-level cost and differentiability, and inject the missing contact physics through a low-dimensional, process-indexed term whose parameters are interpretable and cheap to calibrate (about six calendering states).

Rarefied gas conduction in confined pores. That gas conducts poorly in pores comparable to its molecular mean free path is textbook kinetic theory [16] and is quantitatively central to insulation and aerogel engineering [17], where pore sizes resemble those of battery separators. Coupling this effect into the ZBS framework for battery components appears to be missing from the literature, and the authors of the primary dataset name it as an open issue [4]. Sec. 3.3 supplies the term, and Sec. 7 the experiment that isolates it: gas conduction depends on pressure, skeleton conduction does not.

Battery component thermal measurements. Component-level data come from steady-state stacked-sample rigs with controlled compaction [18, 2], copper-block reference-bar setups on cell teardowns [19, 7], and flash-diffusivity metrology with penetration-model evaluation [21, 4]; canonical early values are due to Maleki et al. [20]. Two robust experimental findings frame our work: soaking a coating with electrolyte raises its conductivity roughly threefold [18, 19], and interfaces, not bulk layers, dominate stack resistance [7]. The first is a clean test of any closure’s fluid sensitivity; the second both motivates our contact term and warns that stack-derived literature values are lower bounds for single coatings.

3 Model

3.1 Setting, notation, and assumptions

We model a porous coating as a two-phase medium: solid particles of conductivity λ_s and diameter d_p , and a pore fluid of conductivity λ_f (a gas or the liquid electrolyte), at porosity ε (the pore volume fraction). The closure predicts the dimensionless ratio $k_{\text{bed}} = \lambda_{\text{eff}}/\lambda_f$ as a function of ε and the phase contrast $\kappa = \lambda_s/\lambda_f$. Calendering is indexed by the compression rate

$$\Pi = 1 - \frac{s_{\text{co}}}{s_{\text{co},0}}, \quad (1)$$

where s_{co} is the coating thickness after calendering and $s_{\text{co},0}$ the as-coated thickness, so $\Pi = 0$ means uncalendered and $\Pi = 0.5$ means compressed to half thickness. Our assumptions, stated up front:

- A1** Steady diffusive heat conduction only. Radiation and convection inside the pores are negligible at micrometer pore sizes and ambient temperature (we verified the porous-convection threshold is missed by about fifteen orders of magnitude at coating scale).
- A2** The closure’s unit cell assumes a statistically isotropic packing of spheres. Real deviations are not remeshed but absorbed into interpretable parameters: the strong conductivity anisotropy of graphite flakes goes into an effective λ_s that must stay inside the single-crystal anisotropy band, and all solid–solid contact conduction goes into the contact term of Sec. 3.4.

- A3** The pore gas behaves as a continuum except for the conductivity reduction of Eq. (4); the liquid electrolyte needs no such correction because molecular mean free paths in liquids are sub-nanometer.
- A4** A single scalar φ stands for the area fraction of all conductive solid–solid bridges (particle–particle necks, binder/carbon-black bridges, coating–collector contact), and the bridge material may differ from the particles, with its own conductivity λ_b .
- A5** Within one electrode family (one recipe and coating process), the material parameters ($\lambda_s, \lambda_b, \varphi_0, a, b$) are constants; only ε and Π change along the calendaring trajectory.

Why build on ZBS rather than on one of the other closures of Sec. 2? Three reasons guided the choice. First, *morphology*: electrodes really are packings of particles, so ZBS’s unit cell is the right starting picture, whereas Maxwell–Eucken’s isolated particles and Bruggeman’s interpenetrating phases are not (Sec. 5.2 confirms this empirically). Second, *validation headroom*: the base closure was validated in companion work on Zehner’s original packed-bed data across phase contrasts $\kappa \in [4, 14094]$, and the battery systems’ values ($\kappa = 14$ –1005) lie strictly inside that range, so we *interpolate* in κ and extrapolate only in porosity (examined empirically in Sec. 5.5). Third, *extensibility*: ZBS fails on battery components at exactly two nameable points, the pore-gas conductivity and the contact term, and both can be repaired without touching the rest of the model.

3.2 Base closure (M0)

The ZBS closure [8, 9, 10] pictures the bed as two parallel channels: a pure-fluid channel of relative cross-section $1 - \sqrt{1 - \varepsilon}$, and a particle-core channel of cross-section $\sqrt{1 - \varepsilon}$ in which heat must pass through fluid gaps between curved particle surfaces:

$$k_{\text{bed}}(\varepsilon, \kappa) = \underbrace{1 - \sqrt{1 - \varepsilon}}_{\text{fluid channel}} + \underbrace{\sqrt{1 - \varepsilon} \lambda_{pm}(\varepsilon, \kappa)}_{\text{particle-core channel}}. \quad (2)$$

The core term λ_{pm} results from integrating the local series resistance across the unit cell:

$$\lambda_{pm} = \frac{2}{1 - B/\kappa} \left[\frac{B(\kappa - 1)}{\kappa(1 - B/\kappa)^2} \ln \frac{\kappa}{B} - \frac{B + 1}{2} - \frac{B - 1}{1 - B/\kappa} \right], \quad B = C \left(\frac{1 - \varepsilon}{\varepsilon} \right)^{10/9}, \quad (3)$$

where B is the particle deformation parameter (it controls how strongly the unit-cell particle is flattened) and C a shape factor, $C = 1.25$ for spheres. The structure of Eq. (2) is worth a moment of reflection, because it explains the model’s behavior later. In the limit $\varepsilon \rightarrow 1$ the fluid channel takes over and $k_{\text{bed}} \rightarrow 1$, as it must. For large phase contrast κ , the core term grows only logarithmically, because the bottleneck is the thin fluid gap where the particles approach each other: no matter how conductive the particles are, heat must squeeze through that gap. This is precisely why the model *under*-predicts whenever real coatings provide solid bridges across the gap, and why a contact term is the physically correct repair rather than, say, inflating λ_s . Equations (2)–(3) with continuum λ_f constitute our baseline M0.

3.3 Knudsen-corrected pore gas

Gas conducts heat by molecules colliding and passing kinetic energy along. When the pore becomes comparable to the mean free path, the average distance a molecule travels between collisions, a molecule crosses the pore and hits the wall before it ever meets another molecule: the pore gas stops behaving like a heat-carrying medium and becomes a lonely messenger bouncing down an empty corridor. The standard correction (often attributed to Smoluchowski) writes the effective gas conductivity as

$$\lambda_{g,\text{eff}} = \frac{\lambda_g}{1 + 2\beta \text{Kn}}, \quad \text{Kn} = \frac{\Lambda}{d_{\text{pore}}}, \quad \Lambda = \frac{k_B T}{\sqrt{2} \pi d_g^2 p}, \quad (4)$$

where λ_g is the continuum gas conductivity, Kn the Knudsen number, Λ the mean free path, d_{pore} the pore diameter, k_B the Boltzmann constant, T the absolute temperature, p the gas pressure, d_g the effective molecular collision diameter (so $\Lambda_{\text{air}} \approx 67$ nm and $\Lambda_{\text{He}} \approx 190$ nm at ambient conditions), and β a dimensionless jump coefficient of order unity that bundles the gas–surface energy accommodation; we use $\beta = 1.64$, a standard value for air on technical surfaces, and quantify the sensitivity to this choice in Sec. 5.5. Two limits anchor the formula: for $\text{Kn} \rightarrow 0$ (large pores) the continuum value is recovered, which

is why packed beds of millimeter particles never needed this correction; for $\text{Kn} \rightarrow \infty$ the gas conduction vanishes entirely. The pore size follows the hydraulic estimate for sphere packings,

$$d_{\text{pore}} = \frac{2}{3} \frac{\varepsilon}{1 - \varepsilon} d_p, \quad (5)$$

or is taken from porosimetry where available (polyolefin separators: 43–64 nm). The corrected contrast $\kappa_{\text{eff}} = \lambda_s / \lambda_{g,\text{eff}}$ then enters Eq. (2) unchanged. Note the practical subtlety this introduces: the “dry” conductivity of a coating now depends on *which gas* fills the pores and at *what pressure*, so a laser-flash measurement under helium purge probes a different pore-gas state than a coating sitting in air, and both differ from a coating under vacuum. Section 7 exploits this pressure dependence as a validation experiment.

3.4 Calendering-aware contact term

Calendering acts on contact zones, not on pore volume, which is the root reason porosity-only models cannot represent its signature. To give the closure eyes for contacts, we adopt the flattened-contact generalization of Eq. (2) from the VDI Heat Atlas [10] and extend it with a bridge phase whose conductivity may differ from the particles’:

$$k_{\text{bed}} = 1 - \sqrt{1 - \varepsilon} + \sqrt{1 - \varepsilon} \left[\varphi \frac{\lambda_b}{\lambda_f} + (1 - \varphi) \lambda_{pm}(\varepsilon, \kappa) \right], \quad (6)$$

where φ is the fraction of the particle-channel cross-section occupied by solid bridges and λ_b is the conductivity of whatever forms those bridges: graphite–graphite flake contacts ($\lambda_b \approx 130 \text{ W m}^{-1} \text{ K}^{-1}$) in anodes, the carbon-black/binder network ($\lambda_b \approx 24 \text{ W m}^{-1} \text{ K}^{-1}$, the carbon-black value cited in [4]) between NMC particles in cathodes. The product $\varphi \lambda_b$ is the physically meaningful group, an areal bridge conductance: doubling the bridge area at half the bridge conductivity leaves the heat flow unchanged, and Sec. 5.5 shows the data indeed determine this product more robustly than its factors. Setting $\varphi = 0$ recovers M0 exactly, a property we enforce in the code by regression test.

How should φ depend on calendering? The dataset’s own adhesion measurements give the answer: Gandert et al. [4] observe that coating–collector adhesion first *drops* with calendering (the roll gap shears the coating and damages the as-coated binder network) and then *recovers* at high line loads (particles interlock and even press into the soft aluminum foil, which they document by electron microscopy). Calendering, in short, first burns the bridges it later rebuilds. The lowest-order polynomial able to express a dip followed by a recovery is a quadratic, so we set

$$\varphi(\Pi) = \max(0, \varphi_0 + a \Pi + b \Pi^2), \quad (7)$$

with three nameable coefficients: $\varphi_0 > 0$ is the as-coated bridge fraction (expected near the rigid-sphere contact value of 0.0077 tabulated in the VDI Heat Atlas), $a < 0$ is the shear-damage rate, and $b \geq 0$ is the interlocking recovery, a threshold effect that a gentle process window may simply never reach. We considered and rejected richer forms: a piecewise-linear $\varphi(\Pi)$ adds a breakpoint parameter that the six to eight data points per family cannot support, and an exponential decay-plus-growth form fits no better while hiding the dip location inside a nonlinear combination of rates. The quadratic is the most parsimonious shape that can be falsified by the adhesion correlation proposed in Sec. 7. We write M1 for Eqs. (6)–(7) restricted to constant φ ($a = b = 0$) and M2 for the full quadratic.

3.5 Differentiable implementation and inverse use

Equations (2)–(7) are implemented in JAX with 64-bit precision, which means every derivative of the model is available exactly, by automatic differentiation, rather than by finite differences with their step-size headaches. The inverse map for porosity quality control is Newton’s iteration

$$\varepsilon_{k+1} = \varepsilon_k - \frac{\lambda_{\text{eff}}(\varepsilon_k) - \lambda^{\text{meas}}}{\partial \lambda_{\text{eff}} / \partial \varepsilon(\varepsilon_k)}, \quad (8)$$

where λ^{meas} is the measured conductivity and ε_k the porosity estimate at iteration k ; the iteration converges quadratically because the derivative is exact, and the solution is unique because $\lambda_{\text{eff}}(\varepsilon)$ is strictly monotonic over the relevant porosity range (verified numerically; intuitively, more pores always means less conduction once the contact state is fixed). First-order uncertainty propagation turns measurement and model noise into a porosity error bar:

$$\sigma_\varepsilon = \frac{\sqrt{\sigma_{\text{meas}}^2 + \sigma_{\text{model}}^2} \lambda_{\text{eff}}}{|\partial \lambda_{\text{eff}} / \partial \varepsilon|}, \quad (9)$$

where σ_{meas} is the relative noise of the conductivity measurement (laser flash or transient plane source: typically 2–5%) and σ_{model} a relative model-form floor, which we set to 2%, the residual scatter of the base closure’s packed-bed validation. The formula also explains *which* electrode makes the best QC target: the larger the sensitivity $|\partial\lambda_{\text{eff}}/\partial\varepsilon|$, the smaller the porosity uncertainty, and the anode has the largest sensitivity of the three components.

4 Data, baselines, and calibration protocol

4.1 Primary dataset

Gandert et al. [4] (open access, CC-BY) measured two graphite anodes on copper foil (96 wt% active material, CMC/SBR binder, 1.5% carbon black; two coating thicknesses) and NMC622/NMC811 cathodes on aluminum foil (PVDF binder; NMC622 additionally 2 wt% flake-graphite additive), each calendered on a laboratory roll calender to six to eight states. Conductivity follows from $\lambda = \kappa_d \rho c_p$, where κ_d is the laser-flash thermal diffusivity (measured under *helium* purge with penetration-model evaluation [21]), c_p the specific heat from differential scanning calorimetry, and ρ the density from gas pycnometry; thicknesses come from a flat-anvil micrometer at contact pressures below 0.3 N mm^{-2} (the authors show that ball-tip dial gauges compress the coating and read up to 23% low); porosity uncertainties follow the Guide to the Expression of Uncertainty in Measurement. We transcribe their per-sheet porosities and thicknesses and evaluate their published fit functions at those porosities. The metal foil is removed by series subtraction,

$$\frac{s_{\text{co}}}{\lambda_{\text{co}}} = \frac{s_{\text{stack}}}{\lambda_{\text{stack}}} - \frac{s_{\text{cc}}}{\lambda_{\text{cc}}}, \quad (10)$$

where subscripts co, stack, and cc denote coating, coated stack, and current collector, s are thicknesses and λ conductivities ($\lambda_{\text{Cu}} = 400$, $\lambda_{\text{Al}} = 237 \text{ W m}^{-1} \text{ K}^{-1}$; the foil term is below 0.1% of the total). One systematic remains by construction: the coating/collector contact resistance stays lumped inside λ_{co} , exactly as in the source. Because the flash measurements run under helium, all dry comparisons use helium as the pore gas ($\lambda_g = 0.1518 \text{ W m}^{-1} \text{ K}^{-1}$), and since $\Lambda_{\text{He}} \approx 190 \text{ nm}$ is almost three times the air value, this dataset stresses the Knudsen term roughly three times harder than air would: an accident of metrology that works in the validation’s favor.

4.2 Anchor datasets

Independent anchors, compiled with row-level citations in the repository: dry polyolefin separators $0.07\text{--}0.18 \text{ W m}^{-1} \text{ K}^{-1}$ across types and compaction pressures [2]; a ceramic-coated separator stack at 0.10 dry and 0.11 wet [19]; soaked/dry electrode pairs, graphite $0.30 \rightarrow 0.89$ and LCO $0.36 \rightarrow 1.10 \text{ W m}^{-1} \text{ K}^{-1}$ [18], CMS graphite $0.57 \rightarrow 1.35$ and LMO $0.16 \rightarrow 0.45 \text{ W m}^{-1} \text{ K}^{-1}$ [19]; the interface-dominance result of [7]; separator pore sizes (43–64 nm) from porosimetry. Constituent properties used throughout: $\lambda_{\text{air}} = 0.026$, $\lambda_{\text{electrolyte}} = 0.18 \text{ W m}^{-1} \text{ K}^{-1}$ (carbonate-solvent class [20]), particle sizes $d_p = 18 \mu\text{m}$ (graphite) and $10 \mu\text{m}$ (NMC).

4.3 Baselines

We compare against the classical closure hierarchy, all evaluated with the *same* constituent properties and the same Knudsen-corrected pore gas, so that differences isolate the geometric assumptions rather than the inputs. The Wiener bounds [11] are the parallel and series arrangements. Maxwell–Eucken [13, 14], for isolated solid spheres of volume fraction $1 - \varepsilon$ dispersed in connected fluid, reads

$$\lambda_{\text{eff}}^{\text{ME}} = \lambda_f \frac{\lambda_s + 2\lambda_f + 2(1 - \varepsilon)(\lambda_s - \lambda_f)}{\lambda_s + 2\lambda_f - (1 - \varepsilon)(\lambda_s - \lambda_f)}, \quad (11)$$

and Bruggeman’s symmetric effective-medium theory [15], which treats both phases as interpenetrating, defines λ_{eff} implicitly as the root of

$$\varepsilon \frac{\lambda_f - \lambda_{\text{eff}}}{\lambda_f + 2\lambda_{\text{eff}}} + (1 - \varepsilon) \frac{\lambda_s - \lambda_{\text{eff}}}{\lambda_s + 2\lambda_{\text{eff}}} = 0. \quad (12)$$

The two microstructure-resolved literature models are not re-implemented; their failure on this dataset (monotonic in porosity, no u-shape) is documented by its authors [4].

4.4 Calibration, ablation, and error-analysis protocol

Per electrode family, the parameter vector $\theta = (\lambda_s, \varphi_0, a, b)$ minimizes the sum of squared *relative* residuals,

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_i \left[\frac{\lambda_{\text{eff}}(\varepsilon_i, \Pi_i; \theta) - \lambda_{\text{co},i}^{\text{meas}}}{\lambda_{\text{co},i}^{\text{meas}}} \right]^2, \quad (13)$$

solved by trust-region least squares. Relative residuals are a deliberate choice: the families span conductivities from 0.5 to 2.4 W m⁻¹ K⁻¹, and absolute residuals would let the anodes dominate the fit while the cathode sheets degenerate into spectators. The parameter box Θ encodes prior physical knowledge rather than convenience: $\lambda_s \in [5, 139]$ W m⁻¹ K⁻¹ for graphite (the single-crystal anisotropy band, from cross-plane to in-plane) and $[1.5, 5]$ for NMC; $\varphi_0 \in [0, 0.08]$; $a \in [-0.2, 0.3]$; $b \in [0, 0.8]$. The accuracy metric is the mean absolute percentage error (MAPE) over all sheets of a family. The protocol comprises four honesty checks: (i) an *ablation* M0→M1→M2, so that the contribution of each model component is assessed separately; (ii) *held-out transfer*, calibrating on graphite_{thin} and predicting graphite_{thick} (same recipe, doubled coating thickness), and calibrating on NMC622 and predicting NMC811 (different additive recipe); (iii) a *profile likelihood* over λ_s , refitting the remaining parameters on a grid of fixed λ_s values, to expose what the data cannot determine; (iv) *hybrid swaps* that attribute transfer error to individual parameters.

5 Results and analysis

5.1 Knudsen magnitudes and zero-fit reference values

Figure 1 quantifies contribution C1. At reference states ($\varepsilon = 0.30$ for electrodes, 0.40 for the separator) the pore-gas conduction loss in air is 4.3% for the graphite anode ($d_{\text{pore}} = 4.9 \mu\text{m}$), 8.8% for NMC (2.3 μm), and 84% for the separator at its measured 43–64 nm pores, where $\text{Kn} = 1.0$ –1.6 means the mean free path *exceeds* the pore size. Propagated to the dry coating conductivity, the electrode corrections are –3.4% and –5.8%; under the helium atmosphere of flash metrology they grow to 2.8–5.5%, which is comparable to the reproducibility of the measurement itself and therefore worth carrying, and we supply it here for exactly the configuration in which the primary dataset was acquired. Zero-fit reference values (Fig. 2) land inside published ranges with no electrode-specific fitting: anode 2.56 (wet) and 0.60 (dry); cathode 0.97 and 0.31; separator 0.29 and 0.065 W m⁻¹ K⁻¹ (the dry value would read 0.116 if the Knudsen term were wrongly omitted). The model’s predicted wet-to-dry ratios (4.3× anode, 3.2× cathode) modestly exceed the measured ones (2.4–3.1× [18, 19]), and the direction of this miss is informative: a real coating contains solid bridges that conduct regardless of the pore fluid, damping its fluid sensitivity, and M0 has no such bridges yet. The ablation will confirm this reading. Porosity sensitivity is large: a routine ± 0.02 calendering tolerance moves electrode λ_{eff} by 5–9% (computed from the exact gradients), so a cell model that treats λ_{eff} as constant embeds a porosity-correlated error of that size.

5.2 Baseline comparison: morphology is the discriminator

Table 1 evaluates the closure hierarchy at the four uncalendered points, the cleanest test because no calender has yet altered any contacts, with identical inputs and Knudsen-corrected helium pore gas throughout. The Wiener bounds are uninformative at graphite’s phase contrast, spanning 0.24 to 32 W m⁻¹ K⁻¹. More interesting is that the structure-committed baselines fail in *opposite, morphology-diagnostic* directions. Bruggeman’s co-continuous picture effectively hands the solid phase a percolating highway through the medium, so it overshoots wherever the solid is actually granular: +17% for NMC622 and up to +465% for graphite. Maxwell–Eucken isolates every particle in a fluid jacket, so heat must repeatedly cross the poorly conducting gas, and it undershoots everywhere (–11% to –70%). M0, spheres touching at points, is the only zero-parameter closure that is both well-ordered across families and accurate where its geometry actually holds: +2.7% for NMC811, the electrode with the least additive (4 wt%) and hard, near-spherical particles. Its systematic *under*-prediction elsewhere (–29% NMC622, –36/–39% graphite) is ordered exactly by expected bridge strength (flake-graphite additive in NMC622; soft, plastically self-bridging flakes in the anodes), and along every calendering series the residual walks monotonically from negative to positive: the u-shape is invisible to every porosity-only model, consistent with the documented failure of the two microstructure-resolved models [4].

The same diagnostic logic settles the separator. Measured values (0.07–0.18 W m⁻¹ K⁻¹) fall between our point-contact lower bound (0.022–0.031) and a continuous-skeleton parallel-plus-Knudsen upper bound

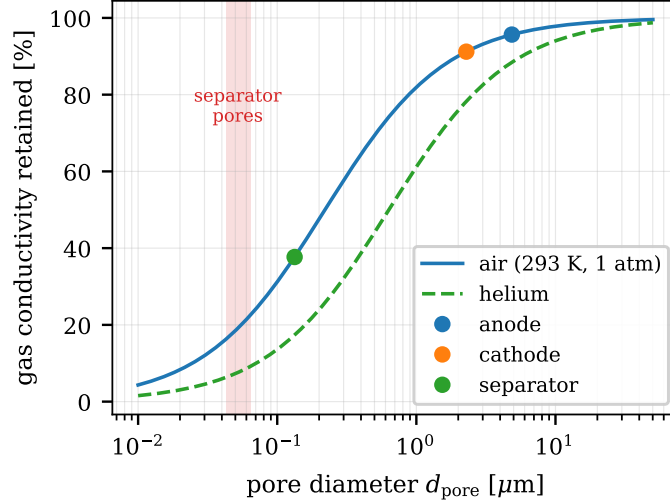


Figure 1: Retained fraction of pore-gas conduction, Eq. (4), at 293 K and 1 atm for air (solid) and helium (dashed). Markers: hydraulic pore sizes of the three reference systems (air curve). Shaded band: measured polyolefin-separator pore diameters; inside it, the air mean free path exceeds the pore size.

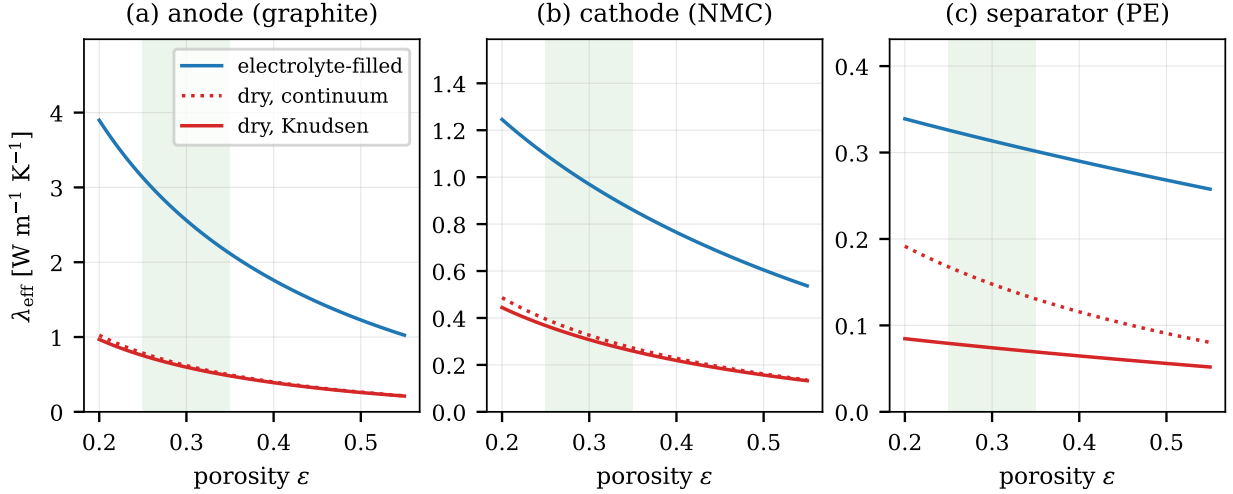


Figure 2: Zero-fit $\lambda_{\text{eff}}(\varepsilon)$ for (a) graphite anode, (b) NMC cathode, (c) PE separator: electrolyte-filled, dry-continuum, and dry-Knudsen variants. Green band: typical post-calendering porosity window.

(0.080–0.155), much closer to the latter, exactly as one expects for a stretched polymer film whose skeleton is co-continuous (Bruggeman, the co-continuous baseline, lands in-range at 0.090). Here a cautionary observation: a continuum sphere-pack model also reproduces the measured numbers (0.075–0.093), but it does so the way a broken clock tells the right time twice a day: it overestimates the gas conduction (no Knudsen correction) while underestimating the skeleton connectivity (wrong morphology), and the two errors cancel. The cancellation is fragile, and it breaks the moment the gas or its pressure changes; the two-pressure experiment of Sec. 7 is designed to break it on purpose.

5.3 Ablation: what each model component buys

Table 2 makes each component justify its existence. Adding a *constant* contact fraction (M1) more than halves the average error, from 31.1% to 13.5%: the as-coated bridge network carries a substantial share of the conduction and cannot be treated as a small correction. Making the contact fraction *process-dependent* (M2, Eq. (7)) reaches 4.5%, the dataset’s own scatter of about 5%, and reproduces the u-shape (Fig. 4): the dip is bridge damage ($a < 0$ in all four families) losing against densification, the upturn is densification plus, where it is reached, interlocking recovery. The recovery term $b > 0$ appears in exactly the two families where interlocking is independently evidenced: NMC622, whose particles visibly penetrate the soft aluminum foil in electron micrographs at the highest line load [4], and the thick anode, which carries the highest line force

Table 1: Baselines at the uncalendered points (dry, helium pore gas with Knudsen correction, identical constituent inputs; coating values derived from [4] via Eq. (10)). W^-/W^+ : Wiener series/parallel bounds; ME: Maxwell-Eucken. All values in $W m^{-1} K^{-1}$.

	measured	W^-	ME	Bruggeman	W^+	M0 (ZBS+Kn)
graphite (thin), $\varepsilon = 0.606$	1.429	0.242	0.430	8.07	31.6	0.871
graphite (thick), $\varepsilon = 0.597$	1.405	0.245	0.441	9.07	32.3	0.903
NMC622, $\varepsilon = 0.513$	0.704	0.258	0.434	0.825	1.29	0.498
NMC811, $\varepsilon = 0.507$	0.492	0.260	0.440	0.841	1.30	0.505

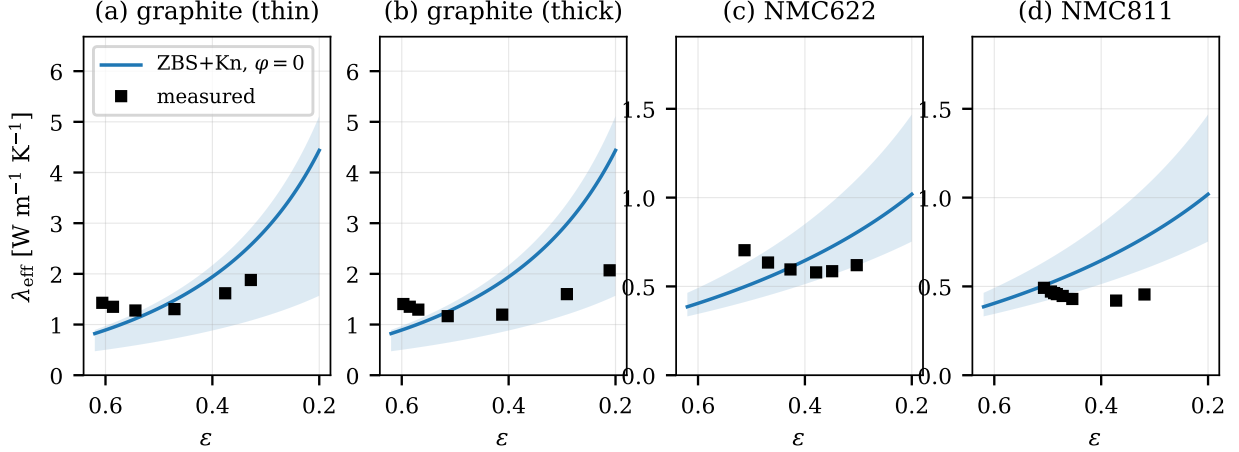


Figure 3: Zero-fit M0 (dry, helium) against the coating conductivities of all 27 calendering states: (a) thin and (b) thick graphite anode, (c) NMC622, (d) NMC811. Bands: λ_s across the physical anisotropy range. The residuals grow systematically toward low porosity; this is the contact signature.

of the four. The fitted parameters are physically legible across the board: $\varphi_0 = 0.005$ – 0.017 brackets the VDI rigid-sphere contact fraction of 0.0077 [10], and the fitted λ_s lands in the low, cross-plane-dominated end of graphite’s anisotropy band, which is what through-plane transport across lying flakes demands. Held-out transfer completes the picture: within a recipe, the calibration carries over usefully (20.9% on the thick anode it never saw); across additive recipes it fails (39.5%, NMC622 to NMC811), and the next subsection dissects why.

5.4 Design principles from the fitted parameters

The fitted parameters are measurements of contact physics, and comparing them across recipes yields design rules, provided the comparison is done in the right units. Binder and additive densities (1.6 – $1.8 g cm^{-3}$) are far below those of the active materials (NMC: 4.7), so a recipe quoted in weight percent hides how much *space* the inactive phase occupies; Table 3 therefore compares in volume. Three interplays emerge. (i) *Bridge conductance*. The as-coated group $G = \varphi_0 \lambda_b$, normalized per unit inactive volume, is roughly $20\times$ larger for the anodes than for the cathodes: soft graphite flakes deform plastically and bridge each other with high-conductivity contacts all by themselves, whereas hard NMC spheres depend entirely on the additive network between them. Within the cathodes, two weight-percent of flake graphite doubles NMC622’s bridge effectiveness over carbon-black-only NMC811: the single clearest recipe lever in the data. (ii) *Damage rate*. The fractional loss $|a|/\varphi_0$ orders perfectly by binder mechanics: stiff PVDF networks

Table 2: Ablation: MAPE over all calendering states. M0: $\varphi = 0$, mid-band λ_s . M1: constant φ , fitted (λ_s, φ_0). M2: full $\varphi(\Pi)$, fitted ($\lambda_s, \varphi_0, a, b$). Right: M2 parameters.

Family	M0	M1	M2	M2 parameters ($\lambda_s, \varphi_0, a, b$)
graphite (thin)	27.3%	9.7%	1.8%	(24.8, 0.0094, -0.024 , 0)
graphite (thick)	48.6%	17.0%	5.4%	(5.0, 0.0121, -0.042 , 0.053)
NMC622	19.1%	14.2%	1.4%	(1.6, 0.0171, -0.089 , 0.039)
NMC811	29.3%	12.9%	9.4%	(1.5, 0.0048, -0.049 , 0)
average	31.1%	13.5%	4.5%	

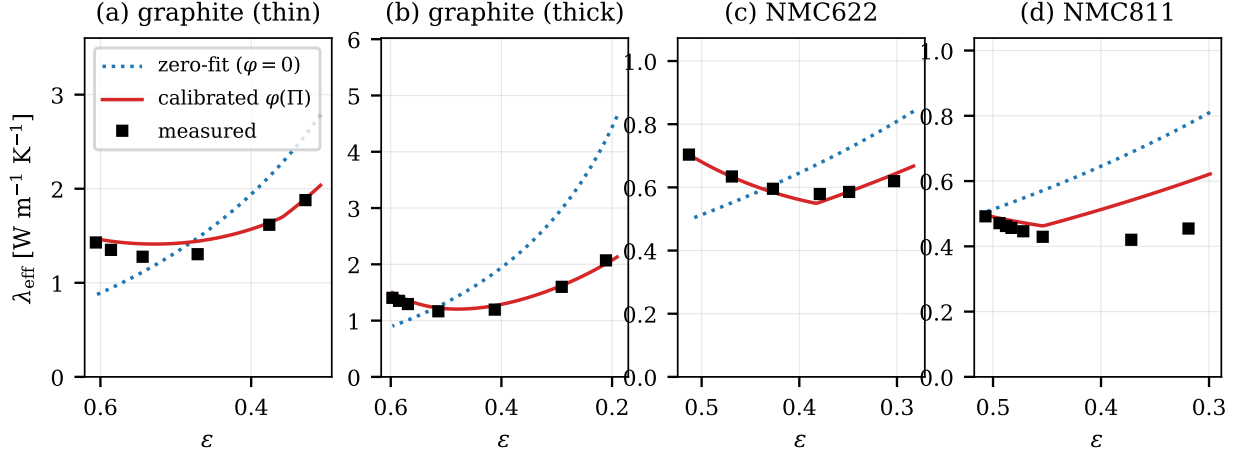


Figure 4: Calibrated M2 closure (solid) vs. M0 (dotted) along each family’s calendaring trajectory; squares: measured coating values. The process-dependent contact term reproduces the u-shape that porosity-only closures cannot.

Table 3: Volume-space decomposition of the contact parameters.

Family	binder+additive [vol% of solid]	$G = \varphi_0 \lambda_b$ [W m ⁻¹ K ⁻¹]	G/vol	$ a /\varphi_0$ [II ⁻¹]	binder
graphite (thin)	6.4	1.22	19.0	2.6	CMC/SBR
graphite (thick)	6.4	1.57	24.4	3.5	CMC/SBR
NMC622	17.8	0.41	2.3	5.2	PVDF
NMC811	10.0	0.12	1.2	10.2	PVDF

fail 2–4× faster per unit compression than elastomeric CMC/SBR, whose rubbery bridges stretch where PVDF’s fracture. *(iii) Recovery* appears only once a process threshold is crossed: NMC811’s $b = 0$ reflects a calendaring window that never reached interlocking ($\text{II} \leq 0.26$), rather than an absence of the underlying mechanism. The design rules: a small flake-graphite additive outperforms additional carbon black by about a factor of two per unit volume; elastomeric co-binders protect the thermal and adhesion network in heavily calendared designs; and the calendaring setpoint should not sit in the conductivity/adhesion dip, so either calender lighter or push through into interlocking.

5.5 Error analysis: identifiability and transfer

What the data can and cannot determine. With four parameters fitted to 6–8 points per family, one must ask which parameters the data actually pin down. Figure 5a answers by profile likelihood: fixing λ_s anywhere in $[5, 30]$ W m⁻¹ K⁻¹ and refitting the rest keeps the error within two points of optimal, with φ_0 compensating between 0.0073 and 0.0120; beyond $\lambda_s \approx 40$ the fit degrades beyond repair. The data therefore reject the high, in-plane-dominated end of the anisotropy band, consistent with through-plane transport crossing the flakes’ weak axis. At the same time, parameter values inside the valley compensate each other, so we report the grouped quantities of Table 3 together with the valley bounds as the parameter uncertainty, rather than individual point estimates.

Why cross-recipe transfer fails. Figure 5b dissects the 39.5% transfer failure by hybrid swaps, replacing one parameter group at a time in NMC622’s calibration with NMC811’s own value. Swapping λ_s recovers almost nothing (36.2%). Swapping φ_0 alone recovers about 90% of the recoverable error (12.2%, against the 9.4% self-fit floor); adding the damage dynamics (a, b) contributes a further 0.3 points. The conclusion is sharp: cross-recipe failure is recipe physics concentrated in a single parameter, the as-coated bridge conductance set by the conductive-additive system, and not a particle-property or dynamics issue. The practical corollary costs one measurement: re-anchor φ_0 on one as-coated sheet when changing recipes, and the rest of the calibration travels.

Remaining exposures, quantified. The Knudsen jump coefficient $\beta \in [1.5, 2.0]$ moves dry electrode predictions by at most 1.7% but the dry separator by 11.7%; the proposed two-pressure experiment deter-

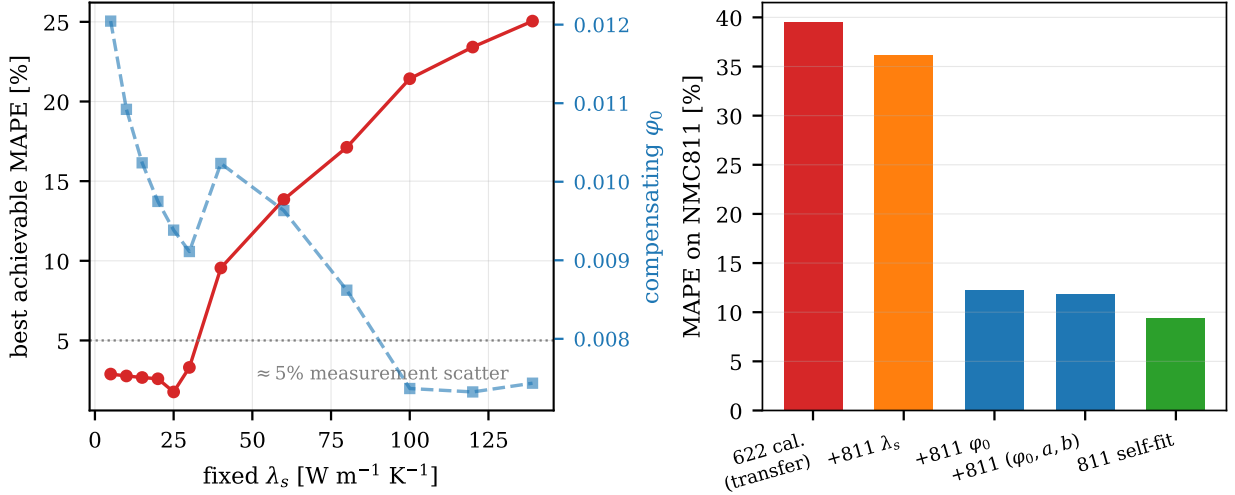


Figure 5: (a) Profile likelihood over λ_s for the thin anode: the λ_s/φ_0 identifiability valley; right axis, the compensating φ_0 . (b) Hybrid-swap decomposition of the NMC622→NMC811 transfer: the as-coated bridge term φ_0 carries the failure.

mines β directly for the separator case. Six of 27 sheets lie below the porosity range of the base closure’s packed-bed validation, with no visible degradation of the calibrated residuals (the extrapolation is absorbed by $\varphi(\Pi)$, an empirical rather than mechanistic resolution). NMC811’s recovery term must not be extrapolated beyond its tested compression range. And all coating values inherit the lumped coating/collector contact resistance of stack metrology [7].

5.6 Manufacturing QC applications

Porosity QC. On a synthetic production batch of 200 calendered sheets (true porosities drawn from $\mathcal{N}(0.30, 0.012)$, a realistic calender drift, with 3% relative measurement noise), Newton inversion (Eq. (8)) recovers porosity with a root-mean-square error of 0.0078 absolute and zero bias (Fig. 6). The a-priori prediction of Eq. (9), $\sigma_\varepsilon = 0.0092$ including the 2% model floor, brackets the observed scatter: the uncertainty model is calibrated and slightly conservative, as is preferable for quality-control use. Against a typical ± 0.02 porosity specification this yields 87.5% correct single-shot classification, with all errors confined to genuinely borderline sheets; averaging $n = 4$ repeats shrinks the measurement contribution by half and makes the specification window cleanly resolvable.

Interface and delamination QC. The calibrated closure decomposes λ_{eff} into a core contribution and a bridge contribution, and the split is large: in the as-coated state, which is exactly where post-drying adhesion and binder-migration quality control matters, bridges carry between 16% and 66% of the heat. A failure of the bridge network (delamination, binder maldistribution) therefore shifts a thermal reading by 5 to 22 standard deviations of a 3%-noise measurement: even partial degradation is detectable in a single shot. The asymmetry with the previous paragraph is worth stating plainly: porosity can already be monitored inline from thickness and mass loading, so thermal porosity QC serves as a convenient secondary check; *interface quality has no inline monitor at all*, and on this evidence a fast thermal reading is its most direct proxy, consistent with the conductivity–adhesion correlation reported independently [4].

6 Discussion: choosing among approaches

The results support a clear division of labor between modeling approaches. *Analytic closures* remain the right tool whenever evaluations must be cheap, differentiable, or invertible, which covers cell-level thermal simulation, process optimization, and quality control, but under two conditions that this work makes precise. The geometry must match the material: Sec. 5.2 shows that the baselines’ failure pattern is itself the diagnostic, point-contact packing for low-additive electrodes, co-continuity for separators. And the process-dependent contact physics must be represented: by Table 2 it is worth a factor of seven in accuracy, far more than the choice between any two textbook closures. *Microstructure-resolved simulation* is the natural tool for exploring hypothetical microstructures before they exist; its present generators,

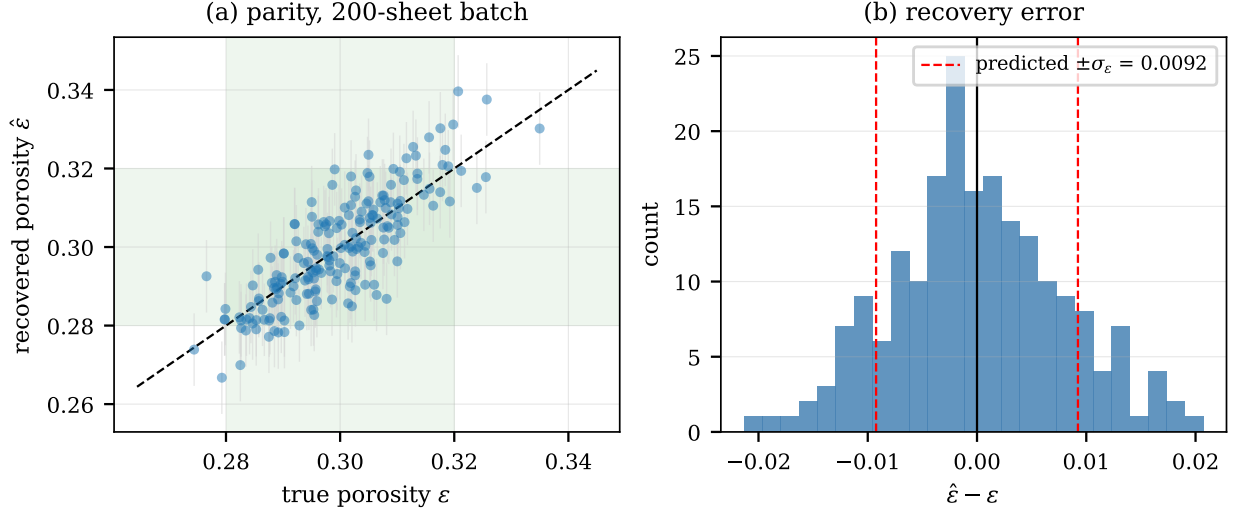


Figure 6: Inverse porosity QC on a synthetic 200-sheet batch at 3% measurement noise: (a) parity with predicted $\pm\sigma_\varepsilon$ bars and the ± 0.02 specification window; (b) recovery-error histogram against the first-order uncertainty prediction of Eq. (9).

however, do not encode calender-induced contact evolution [4], and embedding a $\varphi(\Pi)$ -like contact state into DEM or homogenization generators looks like the obvious synthesis of the two worlds. *Purely data-driven regression* would interpolate the 27 points effortlessly and is the right choice when only interpolation is needed; but it offers no mechanism, no transfer diagnosis when the recipe changes, and no pressure or gas dependence to test. The closure’s value lies in the physical interpretability of its parameters, which converts a fitted curve into design rules (Table 3), a one-measurement transfer correction (Sec. 5.5), and falsifiable predictions (Sec. 7). Seen this way, the per-recipe nature of the contact parameters is itself a quantitative result: contact conduction originates in the contact zones, which bulk volume averages cannot resolve, and six calendering states suffice to characterize it for a new recipe.

7 Proposed experimental validation

The model is calibrated on published data; the following campaign (per recipe: about 26 sheets, one calendering afternoon, two measurement days) is designed to test its *mechanisms*, with stated criteria for failure.

1. **Sample matrix.** Six calendering states from $\Pi = 0$ to the production maximum (single batch, single-side coatings); duplicate as-coated sheets for the φ_0 re-anchor; each state measured dry and electrolyte-soaked.
2. **Porosity ground truth.** Areal mass; flat-anvil micrometer thickness at $\leq 0.3 \text{ N mm}^{-2}$ (not a ball-tip dial gauge, which compresses the coating [4]); helium-pycnometric solid density; GUM uncertainty budget.
3. **Conductivity.** Through-plane laser flash with penetration-model evaluation [21], recording the purge gas (it is the pore gas), or a guarded hot plate; 20°C ; $n = 4$ repeats per sheet, bringing 3% single-shot noise to 1.5%.
4. **Two-pressure Knudsen discrimination** (decisive for separators; recommended for the most porous electrode state): repeat the dry measurement at about 1000 and at or below 100 mbar chamber pressure. By Eq. (4), pore-gas conduction depends on pressure through $\text{Kn} \propto 1/p$, while skeleton and bridge conduction do not. The pressure split therefore isolates the gas channel, pins β , and deliberately breaks the error cancellation identified for continuum sphere-pack separator models in Sec. 5.2.
5. **Adhesion correlation.** Pull-off adhesion strength at every calendering state. Equation (7) claims the conductivity-relevant contact fraction tracks interface damage and interlocking; then $\varphi(\Pi)$ and adhesion must share sign structure and dip location.
6. **Acceptance criteria.** (i) Calibrated residuals at or below the measurement uncertainty at all Π ; (ii) the pressure split consistent with Eq. (4) within the β band; (iii) the $\varphi(\Pi)$ –adhesion correlation

as predicted. Failure of (ii) falsifies the Knudsen term as implemented; failure of (iii) falsifies the contact interpretation and demotes $\varphi(\Pi)$ to a curve fit.

8 Limitations

(1) All conclusions about the contact term rest on one (excellent) public dataset: four families, three recipes, one laboratory, one metrology chain; the campaign of Sec. 7 is required before generality can be claimed. (2) With four parameters on six to eight points per family, λ_s and φ_0 are partially confounded at $\Pi = 0$; we therefore report the profile-likelihood valley and grouped quantities rather than over-interpreting point estimates. (3) The quadratic $\varphi(\Pi)$ is phenomenology anchored to adhesion observations, not derived contact mechanics, and b is unconstrained outside each family’s tested compression range. (4) Stack-derived coating values lump the coating/collector contact resistance; free-standing-film metrology would separate it. (5) Electrode porosities extrapolate the porosity range of the base closure’s packed-bed validation; the extrapolation is handled empirically, not mechanistically. (6) The sphere-pack geometry is wrong for stretched-film separators, so we claim only bounds there, never point predictions. (7) Anisotropy is compressed into one scalar effective λ_s ; in-plane conduction and genuinely anisotropic unit cells are out of scope. (8) Wet-state predictions assume fully saturated pores; partial wetting during electrolyte filling is untreated.

9 Conclusion

A minimal, physically interpretable extension of the ZBS closure, Knudsen-corrected pore gas plus a compression-indexed contact term, closes the gap between packed-bed theory and calendered battery electrodes. It quantifies a rarefaction correction that had previously been noted but not evaluated; it outperforms the classical effective-medium hierarchy with zero fitted parameters; calibrated, it reaches the measurement noise floor (4.5% over 27 calendering states) and reproduces the measured u-shaped conductivity–calendering dependence that defeated previous models. Its fitted parameters function as measurements of contact physics: they decompose into design rules for additive choice, binder mechanics, and calendering setpoint; their identifiability is honestly bounded by profile likelihood; and their transfer behavior localizes all recipe dependence in a single re-anchorable number. Differentiability turns the same closure into manufacturing instrumentation, recovering porosity to ± 0.008 from one thermal reading and offering the first candidate inline monitor for interface quality. The specified two-pressure campaign will test this mechanism. Beyond it, several directions remain open: measurements across additional recipes and coating processes to probe the generality of the contact decomposition; an anisotropic unit cell for in-plane conduction; partial-saturation states during electrolyte filling; the embedding of a $\varphi(\Pi)$ -like contact state into microstructure-resolved structure generators; and, on the application side, the integration of the inverse closure into inline calendering and drying quality control.

Reproducibility statement

The complete project, the JAX implementation of the closure (with a regression test enforcing the exact $\varphi = 0$ reduction to the validated base model), the transcribed datasets with row-level source citations, all calibration scripts with the objective, bounds, and seeds stated in Sec. 4.4, the executed analysis notebooks, a 14-test suite that locks every headline number of this paper against silent drift, and the pipeline that regenerates every figure from raw inputs with no hard-coded values, is openly available at:

<https://github.com/j-stoerk/zehner-electrode-thermal>

Data availability

The transcribed datasets in `data/raw/` of the repository above (the calendering transcription and the separator/electrode literature anchors) carry per-row citations to the primary sources [4, 2, 19, 18, 7]; the primary calendering dataset is published open access by its authors.

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